JAMES DE RAJA

Real-Time Rendering Performance Engineer | Unity | XR Profiling

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SUMMARY

Performance/Systems Engineer with 10+ years in Unity and real-time environments, specializing in GPU/CPU bottleneck analysis, instrumentation, and frame pacing stability.
Built deterministic profiling and diagnostics frameworks (Unity 6 URP + OpenXR) to isolate overdraw, MSAA, submission overhead, and frame timing variance with evidence.
Shipped production systems with an observability-first mindset: metrics, diagnostics, reproducible experiments, and mitigation playbooks.

CORE SKILLS (ATS KEYWORDS)

- Real-Time Performance Optimization
- GPU / CPU Bottleneck Analysis
- Profiling & Diagnostics (Unity Profiler, Frame Debugger)
- XR Frame Timing & Frame Pacing (72/90Hz)
- Unity Rendering Pipelines (URP, Built-in)
- Memory & GC Optimization (C#, Unity)

Tools / Technical: Unity 6, OpenXR, RenderDoc, Android Profiling, Xcode Instruments, Git

Keywords: XR performance, frame pacing, frame timing, rendering optimization, overdraw, MSAA, draw calls, batching/instancing, CPU main thread, render thread, GPU RenderLoop, profiling, performance instrumentation.

SELECTED PERFORMANCE ENGINEERING PROJECTS

XR Performance Stress Lab (Unity 6 URP + OpenXR)
- Built deterministic experiments and instrumentation to classify GPU/CPU bottleneck signatures across overdraw, MSAA, and submission scenarios.
- Measured +7.27ms GPU RenderLoop amplification under overdraw and validated findings via Unity Profiler and Frame Debugger evidence.
- Standardized mitigation mapping by bottleneck type to improve triage speed and decision quality across performance incidents.

Tech: Unity 6, URP, OpenXR, Unity Profiler, Frame Debugger

Profiling Toolkit / Runtime Diagnostics Overlay
- Designed runtime instrumentation for frame timing and variance signals to improve observability during active performance incidents.
- Automated evidence collection conventions to reduce ambiguity and accelerate root-cause analysis with reproducible captures.
- Validated mitigation outcomes against deterministic baselines to prevent regression drift between iterations.

Tech: Unity, C#, Unity Profiler, Frame Debugger, Git

EXPERIENCE

Independent Performance/Systems Engineer (Unity Real-Time) | 2024–Present
- Built observability-first diagnostics for frame timing, frame pacing, and rendering optimization decisions in XR and real-time scenes.
- Diagnosed CPU main thread, render thread, and GPU bottlenecks using deterministic workflows and evidence-backed profiling.
- Standardized mitigation playbooks and handoff artifacts to improve repeatability of performance investigations.

Senior Unity Engineer (Performance + Reliability Focus) | 2014–2024
- Improved stability by instrumenting performance-critical systems and validating fixes with reproducible CPU/GPU profiling.
- Reduced recurring runtime issues through structured diagnostics, measurable triage workflows, and validation gates.
- Led performance optimization efforts for shipped mobile projects with emphasis on frame-time stability and maintainability.

EDUCATION

Formal education details available upon request.